

BDS-3 SMC Encryption — Key Infrastructure Comparison

Existing solutions (SM9 IBE, SM4 Symmetric, CGAN/PVT) vs Proposed Pre-Shared Key AES-128-CBC (This Work)

Criterion	SM9 IBE (Jin et al., 2024)	SM4 Symmetric (IEEE, 2023)	CGAN / PVT-Keyed (MDPI, 2023)	Pre-Shared Key AES-128 (This Work)
Requires internet at time of transmission	✗ Yes — needs the key generation centre (KGC) online to issue identity-based private keys before transmission can proceed.	✗ Yes — secure key exchange channel must be active before the first message is sent.	✓ No — key is derived locally from the satellite PVT data. No external connection needed.	✓ No — key is hardcoded into firmware at setup. Transmission works with zero connectivity.
Requires knowledge of receiver identity	✗ Yes — IBE requires the sender to know the receiver's identity string (e.g. email or ID) to compute the encryption key.	✗ Yes — a pre-shared key must have been exchanged with the specific receiver in advance.	✓ No — key is derived from the transmitter's own PVT. Receiver derives the same key independently.	✓ No — the same hardcoded key is in all authorised terminals and decoders. No identity needed.
Works in cold-start emergency scenario (no prior setup)	✗ No — IBE private key must be obtained from the KGC before the emergency. Cannot be done on the spot.	✗ No — symmetric key must be pre-exchanged with the receiver. Cannot be negotiated during the emergency.	■ Partial — works if both sides implement the same PVT-keying algorithm in advance. Complex synchronisation required.	✓ Yes — as long as the firmware is pre-loaded with the key, the terminal transmits securely from the first press of the button.
External key infrastructure needed	✗ Yes — requires a trusted Key Generation Centre (KGC) server, online and reachable during setup.	✗ Yes — requires a secure out-of-band channel (HTTPS, physical exchange) to distribute the key beforehand.	✓ No — both sides use the satellite signal itself as the entropy source. No third party involved.	✓ No — key lives in firmware. Only the deploying organisation needs to manage it at manufacturing time.
Payload overhead added by encryption	■ ~25 ms authentication latency. Encryption itself adds minimal bits but the key exchange protocol adds message overhead.	■ Comparable to AES — block cipher adds padding overhead. SM4 block size is 128 bits.	✗ Significant — the CGAN model must generate a key per message, adding computational and message overhead.	✓ Exactly 64 bits overhead. Binary payload (64 bits) → AES-128 ciphertext (128 bits). Fits within GSMC 560-bit limit.
Implementable on ESP32 without external libraries	✗ No — SM9 requires a dedicated cryptographic library not available in the standard ESP32 Arduino SDK.	■ Partial — SM4 is a Chinese national standard. Requires a third-party library on ESP32.	✗ No — a CGAN model cannot run on an ESP32. Requires a GPU or server-class compute.	✓ Yes — AES-128-CBC is built into the ESP32's mbedTLS library. No additional libraries or hardware needed.
Protects against plaintext interception of S-band downlink	✓ Yes — coordinates are encrypted. An interceptor cannot decode the position without the private key.	✓ Yes — SM4 encryption prevents plaintext reading by unauthorised S-band receivers.	✓ Yes — PVT-keyed encryption hides coordinates from passive interception.	✓ Yes — AES-128 ciphertext is computationally infeasible to break without the shared key.

Criterion	SM9 IBE (Jin et al., 2024)	SM4 Symmetric (IEEE, 2023)	CGAN / PVT-Keyed (MDPI, 2023)	Pre-Shared Key AES-128 (This Work)
Key rotation / compromise recovery	✓ Strong — IBE private keys can be revoked and reissued by the KGC without changing the public key.	■ Moderate — requires re-distributing a new symmetric key to all terminals. Operationally complex.	✓ Strong — key changes automatically each message as PVT changes. No explicit rotation needed.	✗ Weak — if the hardcoded key is compromised, all terminals must be reflashed with a new key. No remote rotation.
Suitable for mass deployment (1000s of terminals)	✓ Yes — IBE scales well. Each user gets their own identity-based key from the KGC.	■ Difficult — each terminal needs its own securely distributed key or all share one (security risk).	✓ Yes — each terminal derives its own key from its own PVT. No central management.	✗ No — a single shared key across all terminals means one compromised device exposes all. Not suitable for large scale.
Overall suitability for Lab 7 / BDS-SMC2 experiment	✗ Not suitable — requires KGC infrastructure unavailable in a lab setting.	✗ Not suitable — requires pre-arranged key exchange with a specific receiver.	✗ Not suitable — CGAN cannot run on ESP32; too complex for a lab implementation.	✓ Fully suitable — works out of the box on ESP32, no infrastructure, no internet, immediate encryption from first transmission.

✓ Advantage

✗ Disadvantage

■ Partial / conditional

Green column = Proposed solution (This work)